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**Sardar Patel Institute of Technology**
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**Department of Computer Engineering**

## Bachelor of Technology

## Major Project – Phase I Report

**Academic Year: 2025–2026**

# MentorLink

*A Data-Driven Interaction and Academic Analytics Platform*

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**Under the Guidance of**

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# Approval Certificate

This is to certify that the Project entitled 'Waste2Worth - Saving Resources, Serving Communities' by Himani Pilankar, Tithi Talele and Chetan Wankhede is approved for the partial fulfillment of Mini Project - I for the Third Year Mini Project towards obtaining the Degree of Bachelor of Technology in Computer Science and Engineering.

**Project Guide**

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**Principal**

**Seal of the Institute**

## Declaration

I wish to state that the work embodied in this work titled "Waste2Worth - Saving Resources, Serving Communities" forms my own contribution to the work carried out under the guidance of Prof. Soni Bhambar at the Sardar Patel Institute of Technology. I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission.


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## Abstract

Peer mentorship plays a significant role in academic and professional development within higher educational institutions. Senior students frequently guide junior students in areas such as curriculum planning, internship preparation, project development, and career decision-making. However, in most institutions, mentorship interactions occur informally through personal networks, messaging platforms, or classroom discussions. These interactions lack structured documentation, centralized tracking, and institutional visibility.

An additional limitation of informal mentorship models is restricted accessibility. Students who are socially hesitant or lack strong peer networks may find it difficult to approach seniors for guidance. In the absence of a structured mentor discovery mechanism, many students remain underserved, leading to unequal access to mentorship opportunities.

The absence of a formal mentorship management infrastructure prevents institutions from analyzing engagement patterns and identifying frequently discussed academic domains.

MentorLink is proposed as a centralized, web-based mentorship management platform designed to formalize peer mentorship within the institutional ecosystem. The system introduces verified authentication, structured student profiles, role-based access control, and a defined mentorship lifecycle model. It also supports topic-based interaction logging and academic discussion forums to ensure that mentorship activities are recorded in a structured manner.

Phase I of the project focuses on requirement analysis, system design, database modeling, backend implementation of core modules, and preparation for future analytics integration.

## List of Figures


1. System Architecture Diagram
2. Data Flow Diagram (DFD) Level 0
3. Data Flow Diagram (DFD) Level 2
4. Class Diagram
5. Cross-System Donation Lifecycle Workflow Diagram
6. Donor Workflow Diagram
7. NGO Workflow Diagram
8. Admin Workflow Diagram
9. AI Verification Workflow Diagram
10. Platform Visitor Workflow Diagram


## List of Tables


1. Literature Survey Summary

# Contents


Abstract 3

1 Introduction 7

1.1 Background . . . 7

1.2 Project Overview . . . 7

1.3 Key Contributions . . . 8

1.4 Target Users . . . 8

2 Problem Definition, Scope and Objectives of the Project 9

2.1 Problem Definition . . . 9

2.2 Scope of the Project . . . 9

2.3 Objectives . . . 10

2.3.1 Primary Objectives . . . 10

2.3.2 Secondary Objectives . . . 10

3 Literature Survey 11

3.1 Scope and Methodology . . . 11

3.2 Paper Summaries . . . 11

3.3 Comparative Summary . . . 12

3.4 Gap Analysis . . . 12

4 Analysis 13

4.1 System Architecture Diagram . . . 13

4.2 Use Case Diagram . . . 14

4.3 Class Diagram . . . 15

4.4 Sequence Diagram (Mentorship Request Flow) . . . 16

4.5 Flowchart (Mentorship Lifecycle) . . . 17

5 Design and Methodology 18

5.1 Proposed System Block Diagram . . . 18

5.2 Major Modules of the Proposed System . . . 18

5.3 Algorithmic Techniques Used . . . 19

5.3.1 1. Mentor Recommendation Algorithm . . . 19

5.3.2 2. Mentorship Lifecycle State Machine . . . 19

5.3.3 3. Structured Interaction Logging Model . . . 20

5.4 Design Principles Followed . . . 20

6 Implementation Details 20

6.1 Implementation Progress (Phase I – 25%) . . . 20

6.2 Module-wise Implementation Screenshots . . . 21

6.2.1 1. User Registration and Login . . . 21

6.2.2 2. Student Profile Creation . . . 22

6.2.3 3. Mentorship Request Workflow . . . . . 23
6.2.4 4. Community Discussion Module . . . . . 24
6.2.5 5. MongoDB Database Collections . . . . . 25
7 Technology Stack 26
7.1 Technology Stack Diagram . . . . . 26
7.2 Frontend Technologies . . . . . 26
7.3 Backend Technologies . . . . . 27
7.4 Database Technology . . . . . 27
7.5 Development and Testing Tools . . . . . 27
7.6 Architecture Pattern . . . . . 27
8 Project Plan and Timeline 27
8.1 Work Completed in Phase I . . . . . 27
8.2 Planned Work for Next Semester (Phase II) . . . . . 28
9 References 29

# 1. Introduction

## 1.1. Background

Mentorship plays a significant role in academic growth and professional development within higher educational institutions. Senior students often guide junior students in academic planning, internship preparation, project development, skill enhancement, and career decision-making. Such peer-to-peer mentorship fosters collaborative learning and strengthens academic communities.

However, in most institutions, mentorship interactions occur informally through personal networks, messaging platforms, or casual communication channels. A major limitation of this informal model is accessibility. Students who are introverted, socially hesitant, or lacking strong peer networks often face difficulty in approaching seniors for guidance. In the absence of a structured discovery mechanism and organized communication workflow, many students remain underserved despite having genuine academic needs.

Additionally, informal communication lacks structured documentation. Mentorship discussions are not recorded in an analyzable format, resulting in repetitive queries, fragmented knowledge sharing, and the absence of reusable academic insights.

## 1.2. Project Overview

MentorLink is proposed as a centralized, web-based mentorship and academic networking platform designed to formalize peer mentorship within the institutional ecosystem. The platform introduces a structured digital infrastructure that enables systematic mentor discovery, formal mentorship request workflows, organized interaction logging, and community-based academic discussions.

The system supports multiple user roles including junior students, senior mentors, and administrative moderators. Junior students can explore mentor profiles based on academic attributes, send mentorship requests, participate in structured mentorship conversations, and engage in domain-based communities. Senior students can accept mentorship requests, guide mentees, and actively contribute to academic discussions and community threads.

A key component of the platform is the mentorship lifecycle model, which defines clear states such as request initiation, acceptance, and active mentorship. This structured workflow ensures clarity and accountability in mentorship relationships. The platform also provides one-to-one messaging, group discussions, and public or private academic communities to support collaborative interaction.

Furthermore, the system incorporates a mentor recommendation engine that suggests suitable mentors based on skill similarity, domain interests, and academic compatibility. Interaction logging mechanisms capture structured metadata for mentorship sessions and community par-

ticipation, preparing the system for future analytical processing.

### 1.3. Key Contributions

The primary contributions of MentorLink are as follows:

- Introduction of a structured mentorship lifecycle model.
- Development of a centralized mentor discovery and recommendation mechanism.
- Implementation of Reddit-style academic communities for collaborative discussion.
- Integration of private and group communication features.
- Design of a structured interaction logging system to capture academic metadata.
- Creation of a scalable backend architecture prepared for future data mining integration.

By combining mentorship management, intelligent recommendation, and community-based interaction within a unified platform, MentorLink transforms informal student guidance into an organized and structured academic ecosystem.

### 1.4. Target Users

MentorLink is designed for the following user groups:

- **Junior Students:** Individuals seeking academic mentorship, domain exploration, and peer guidance.
- **Senior Students (Mentors):** Experienced students who provide academic and career guidance.
- **Administrative Moderators:** Users responsible for maintaining platform integrity and moderating content.

The system aims to provide equitable access to mentorship opportunities while enabling structured interaction and organized academic collaboration.

# 2. Problem Definition, Scope and Objectives of the Project

## 2.1. Problem Definition

In most higher educational institutions, peer mentorship occurs through informal and unstructured channels such as personal messaging applications, social media groups, or casual academic interactions. While these platforms enable communication, they lack a structured framework for mentorship discovery, request management, and interaction tracking.

A significant challenge within existing informal systems is limited accessibility. Students who are introverted, socially hesitant, or lacking strong peer networks often struggle to identify and approach suitable mentors. The absence of a transparent mentor discovery mechanism results in unequal access to academic guidance and missed opportunities for structured support.

Furthermore, current communication channels do not provide a centralized platform for academic discussion and domain-based collaboration. Discussions are fragmented across multiple platforms, leading to repetitive queries, scattered knowledge, and the loss of valuable academic insights.

Another limitation is the absence of structured interaction logging. Mentorship sessions and discussions are not recorded in a systematic format, preventing analysis of engagement patterns, subject demand, or mentorship effectiveness. Without structured data collection, student communities are unable to leverage data-driven approaches for improving academic collaboration.

Additionally, there is no intelligent system to recommend suitable mentors based on skills, interests, or domain compatibility. Students often rely on personal familiarity rather than expertise alignment when selecting mentors, resulting in suboptimal mentorship relationships.

Therefore, there is a need for a centralized, structured, and intelligent mentorship and academic networking platform that enables transparent mentor discovery, organized communication, community-driven discussions, and structured interaction logging within an institute-exclusive environment.

## 2.2. Scope of the Project

The scope of MentorLink is confined to the development of an institute-exclusive web-based mentorship and academic networking platform. The system focuses on formalizing peer mentorship and enabling structured academic collaboration among students.

The project includes:

- Secure institute-based user authentication.
- Structured student profile and academic portfolio management.
- Mentor discovery and recommendation mechanisms.

- Mentorship request and lifecycle management.
- One-to-one messaging and group communication.
- Public and private academic communities.
- Structured interaction logging for mentorship sessions.

The system is designed with a scalable backend architecture to support future integration of advanced data mining and analytics techniques. However, large-scale predictive analytics and machine learning models will be implemented in subsequent phases of the project.

## 2.3. Objectives

### *2.3.1. Primary Objectives*

- To provide a verified, institute-exclusive platform for academic mentorship and networking.
- To enable structured interaction between junior and senior students for academic guidance and collaboration.
- To implement a centralized mentor discovery and recommendation mechanism based on skills and domain interests.
- To establish a structured mentorship lifecycle model including request, acceptance, and active mentorship states.
- To develop Reddit-style academic communities for collaborative discussion and knowledge sharing.
- To capture student interactions as structured transactional data for organized record keeping and future analytical processing.

### *2.3.2. Secondary Objectives*

- To integrate one-to-one and group communication features for seamless academic interaction.
- To design a scalable backend architecture capable of supporting future data mining integration.
- To prepare the system for application of data mining techniques such as clustering, association rule mining, and trend analysis in later phases.
- To ensure modular system design allowing future expansion and enhancement.

# 3. Literature Survey

## 3.1. Scope and Methodology

Donation management systems, intelligent recommendation algorithms, OCR-based document verification, AI-driven content moderation, and sustainability-focused impact assessment frameworks are all covered in this review of the literature. Relevant studies were found using academic databases, industry whitepapers, and technical documentation related to contemporary AI models and verification pipelines in order to build a solid foundation for Waste2Worth. Twelve exemplary works were chosen from these sources on the basis of their methodological significance, technological importance, and suitability for practical implementation. Together, the surveyed works guide the platform's architectural design, encourage the addition of automated verification and matching features, and highlight enduring issues like a lack of trust, a lack of transparency, and insufficient lifecycle tracking in the donation ecosystems that are currently in place.By combining developments from various fields, the review provides a convincing defence of Waste2Worth's design choices and identifies the shortcomings that the suggested system seeks to fill.

## 3.2. Paper Summaries

1. **Smith, A., et al. (2020).** *Digital Redistribution Platforms in Urban Settings.* Reviews major donation/redistribution platforms and highlights common issues such as verification delays, donor drop-offs, and inefficient matching. Motivates the need for automated verification and structured workflows.

2. **Chen, L., et al. (2019).** *OCR-Based Document Verification in Service Platforms.* Compares Tesseract with commercial OCR tools and shows how preprocessing improves text extraction. Supports our choice of Tesseract for NGO document verification.

3. **Li, Y., et al. (2021).** *Hybrid Recommendation Systems for Resource Allocation.* Demonstrates that hybrid recommenders (content + history) outperform single-method approaches, reinforcing our multi-factor scoring model.

4. **Garcia, R., et al. (2021).** *Reputation and Rating Mechanisms in Peer Platforms.* Discusses rating reliability, bias reduction and the importance of transparent feedback—concepts relevant for donor/NGO rating.

5. **Singh, V. & Patel, A. (2021).** *Role-Based Access Control in Distributed Web Systems.* Explores JWT, access layers and token expiry strategies, supporting our authentication and RBAC model.

### 3.3. Comparative Summary

| Paper | Focus | Key Relevance |
|---|---|---|
| Smith et al., 2020 | Redistribution platforms | Highlights need for verification and structured matching workflows. |
| Chen et al., 2019 | OCR verification | Supports use of Tesseract and preprocessing in our verification pipeline. |
| Li et al., 2021 | Hybrid recommenders | Motivates multi-factor scoring in our recommendation system. |
| Garcia et al., 2021 | Reputation systems | Justifies feedback and rating workflow in our platform. |
| Singh & Patel, 2021 | RBAC/JWT security | Supports our authentication and role-based access implementation. |

### 3.4. Gap Analysis

The body of existing literature offers useful answers for topics like recommendation algorithms, content safety analysis, and OCR-based document extraction. These solutions, however, hardly ever come together to create a single donation ecosystem framework. The majority of donation platforms cited in previous research rely on manual verification procedures. They frequently lack intelligent matching capabilities and fail to take into account operational constraints such as item urgency, geographic proximity, or NGO pickup capacity. Furthermore, transparency measures like proof of pickup and delivery are rarely included in current systems. The scalability, dependability, and practical efficiency of donation workflows are limited by the lack of an integrated, AI-driven, trust-focused framework.Waste2Worth fills this gap by integrating sustainability metrics, automated NGO verification, and thorough status tracking into a single platform intended to support the circular economy and resource redistribution.

# 4. Analysis

## 4.1. System Architecture Diagram

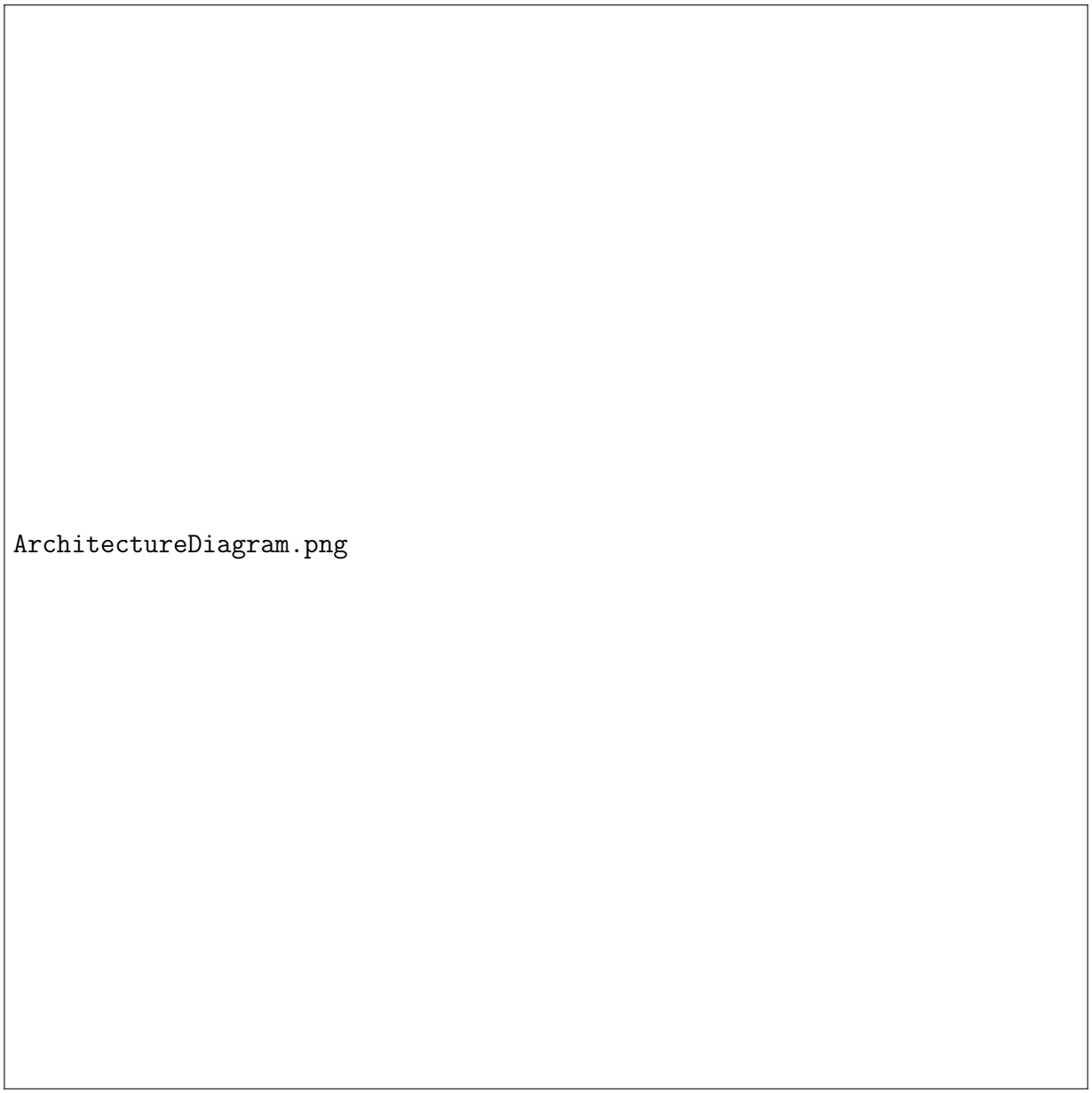



Figure 1: Overall System Architecture of MentorLink

### 4.2. Use Case Diagram

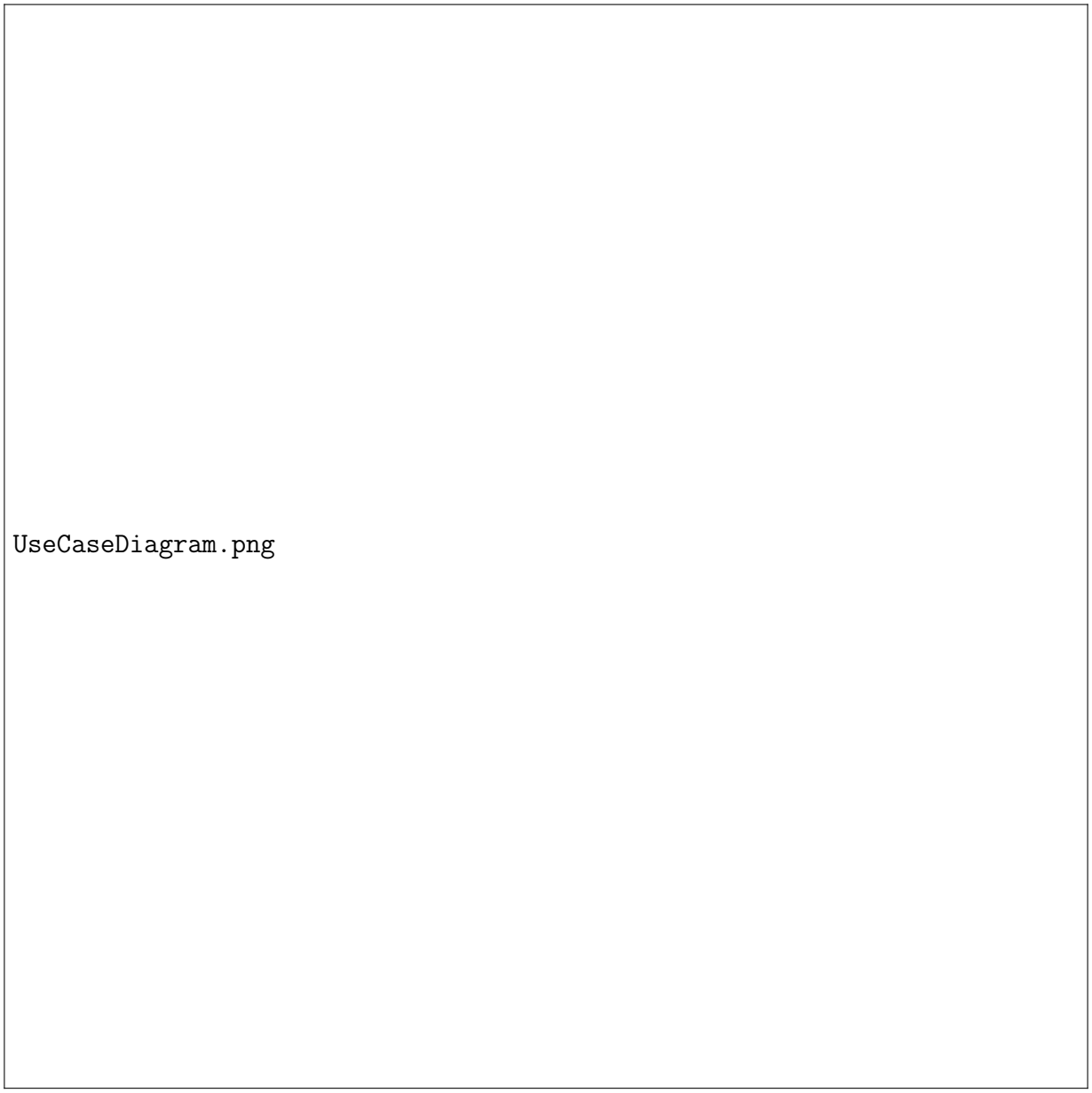



Figure 2: Use Case Diagram Representing System Actors and Interactions

### 4.3. Class Diagram

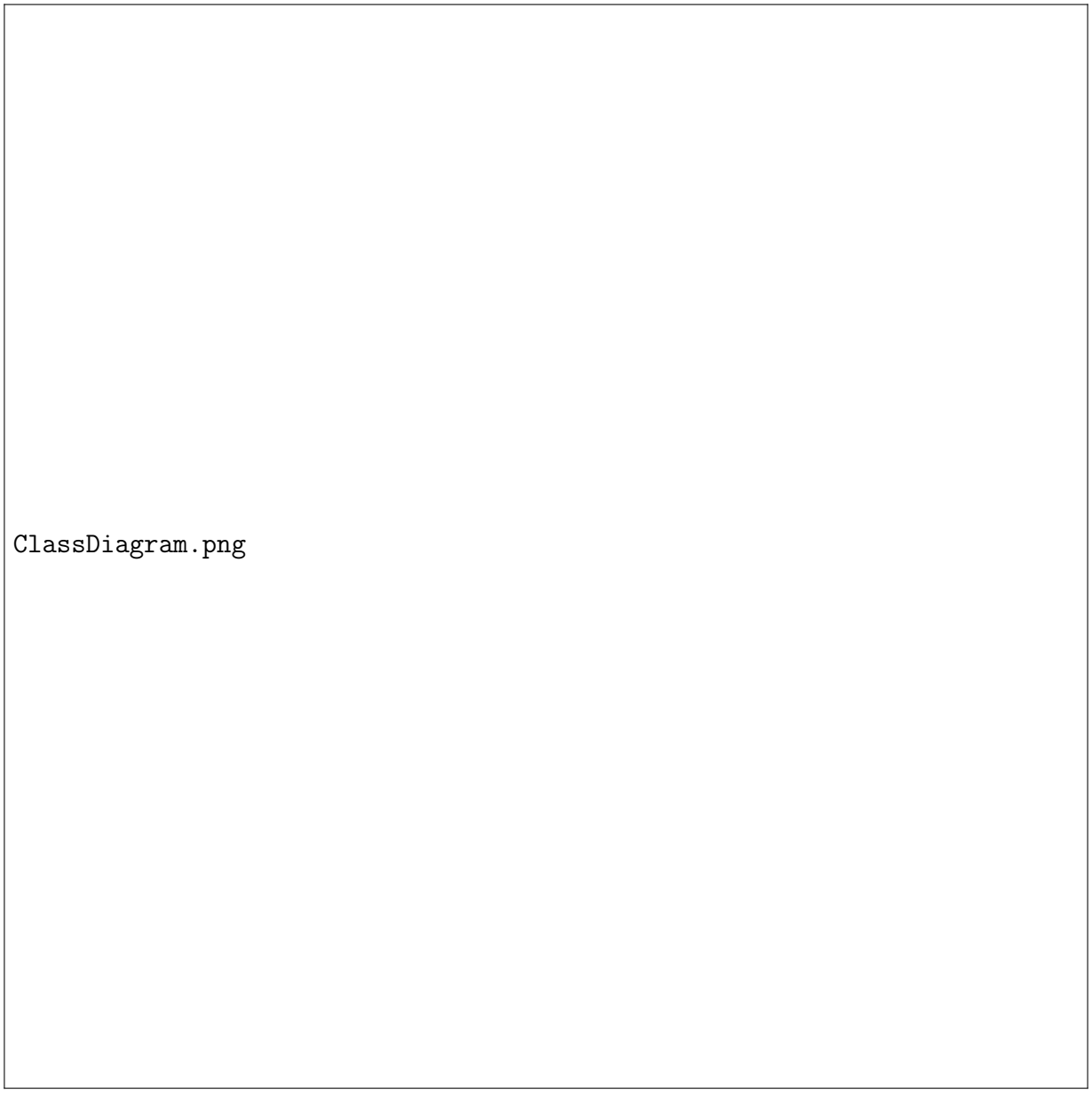



Figure 3: Class Diagram Showing Core Entities and Relationships

### 4.4. Sequence Diagram (Mentorship Request Flow)




Figure 4: Sequence Diagram for Mentorship Request Lifecycle

### 4.5. Flowchart (Mentorship Lifecycle)




Figure 5: Flowchart Representing Mentorship Lifecycle States

# 5. Design and Methodology

## 5.1. Proposed System Block Diagram




Figure 6: Proposed System Block Diagram Highlighting Major Modules

## 5.2. Major Modules of the Proposed System

The MentorLink system is divided into the following major modules:

- **Authentication Module**
- **User Profile Module**
- **Mentorship Management Module**

- **Mentor Recommendation Engine**
- **Community Module**
- **Messaging Module**
- **Moderation Module**
- **Database and Logging Module**

### 5.3. Algorithmic Techniques Used

#### *5.3.1. 1. Mentor Recommendation Algorithm*

The mentor recommendation system follows a weighted similarity scoring approach inspired by content-based filtering techniques used in recommendation systems literature.

$$Score = w_1(SkillOverlap) + w_2(DomainMatch) + w_3(DepartmentMatch) + w_4(YearCompatibility)$$

Where:

- $w_1 = 0.4$
- $w_2 = 0.3$
- $w_3 = 0.2$
- $w_4 = 0.1$

This approach is aligned with similarity-based recommendation models discussed in collaborative filtering research.

#### *5.3.2. 2. Mentorship Lifecycle State Machine*

The mentorship process follows a finite state machine model:

- Requested
- Accepted
- Active
- Closed

State transitions are strictly controlled to ensure system consistency.

#### *5.3.3. 3. Structured Interaction Logging Model*

Each mentorship session is logged with metadata:

- Topic
- Category
- Timestamp
- Duration
- Participants

This structured logging enables future application of data mining techniques such as:

- Clustering
- Association Rule Mining
- Trend Analysis

### 5.4. Design Principles Followed

- Separation of Concerns
- Modular Architecture
- Role-Based Access Control (RBAC)
- Scalability-Oriented Backend Design
- Analytics-Ready Data Structuring

## 6. Implementation Details

### 6.1. Implementation Progress (Phase I – 25%)

Phase I of MentorLink focuses on establishing the foundational backend architecture and core system modules. The following components have been successfully implemented:

- User authentication and registration module.
- Structured student profile creation.
- Community creation and post functionality (basic version).

- MongoDB database schema implementation.

The system is currently functional at the core backend level with partial frontend integration completed.

## 6.2. Module-wise Implementation Screenshots

### *6.2.1. 1. User Registration and Login*

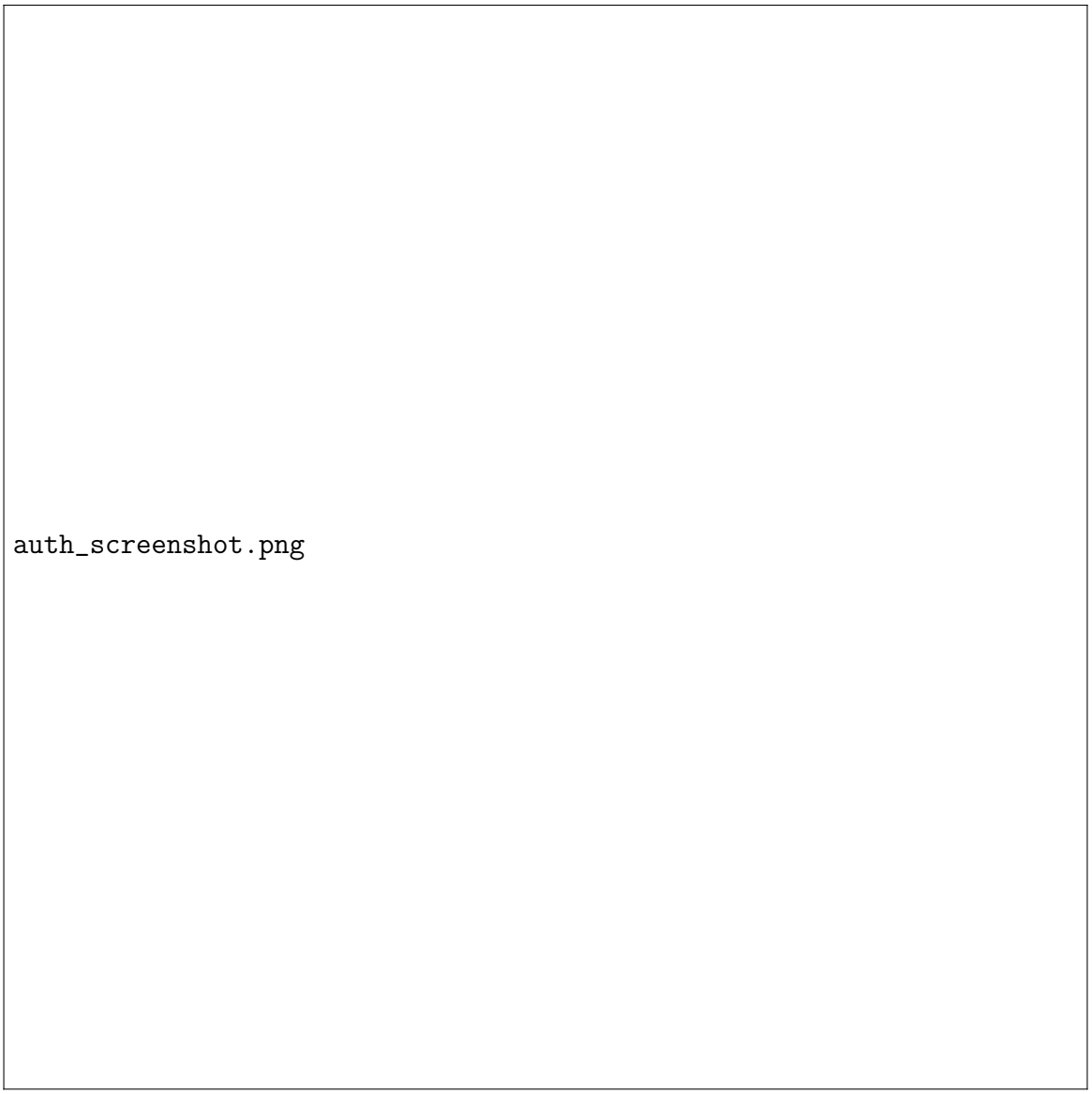



Figure 7: User Registration and Login Interface

*6.2.2. 2. Student Profile Creation*

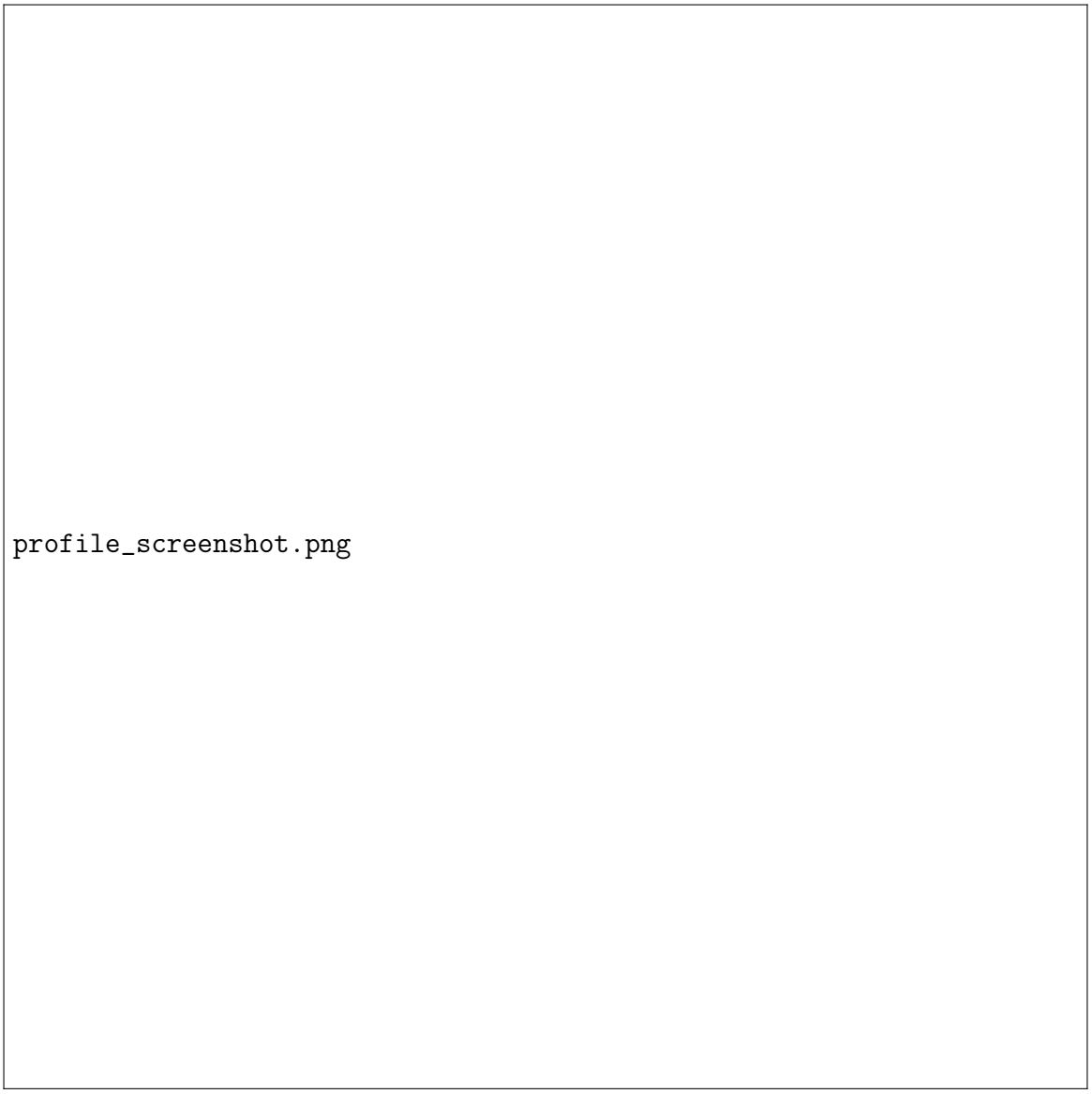



Figure 8: Structured Student Profile Interface

*6.2.3. 3. Mentorship Request Workflow*




Figure 9: Mentorship Request and Acceptance Flow

*6.2.4. 4. Community Discussion Module*




Figure 10: Academic Community and Discussion Posts

*6.2.5. 5. MongoDB Database Collections*




Figure 11: MongoDB Collections for Core Modules

# 7. Technology Stack

## 7.1. Technology Stack Diagram

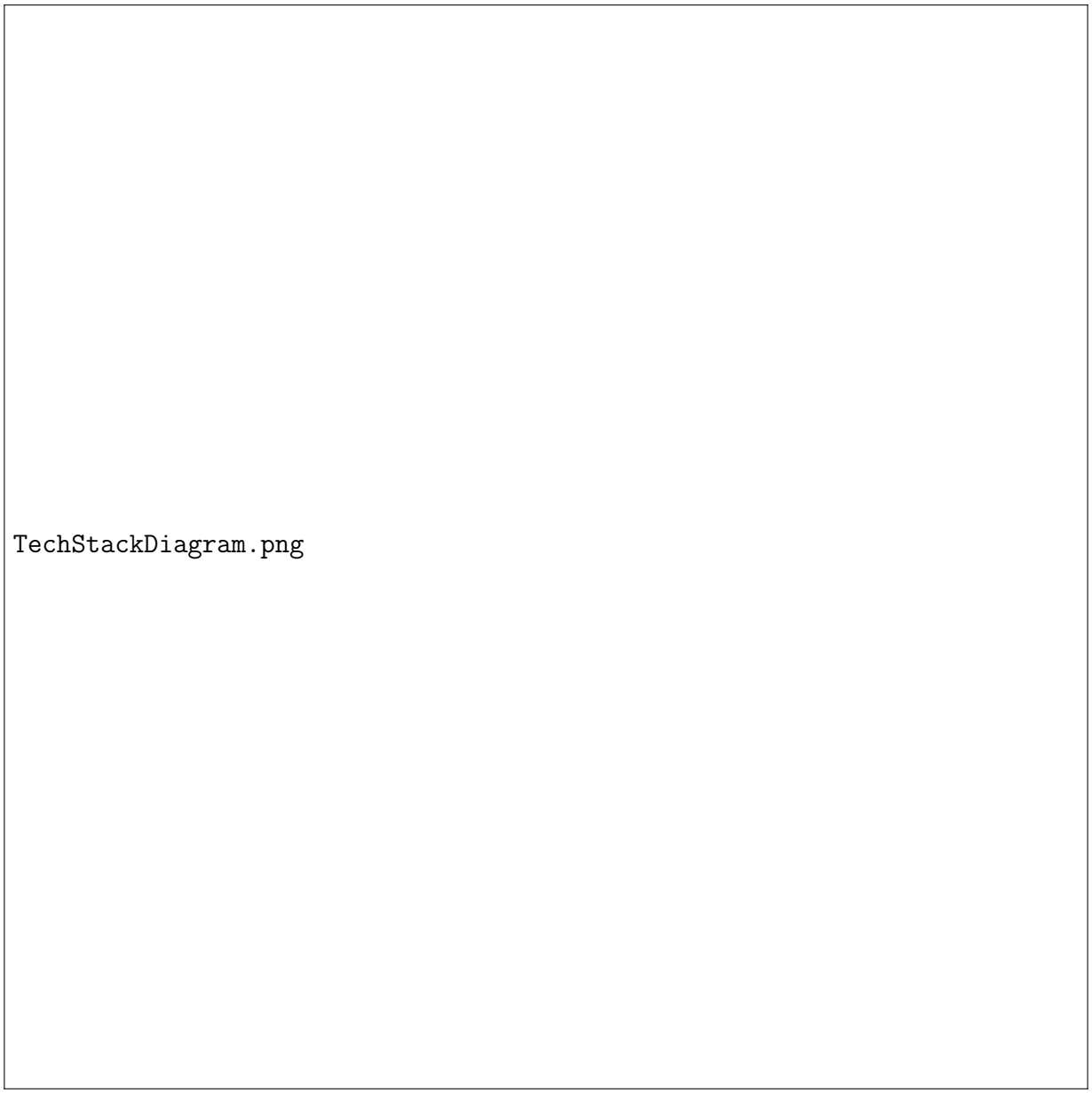



Figure 12: Technology Stack of MentorLink

## 7.2. Frontend Technologies

- **React.js:** Used for building the user interface with a component-based architecture.
- **Vite:** Development and build tool for fast bundling and optimized builds.
- **Tailwind CSS:** Utility-first CSS framework for responsive and consistent UI design.
- **Axios:** Used for HTTP communication between frontend and backend.

### 7.3. Backend Technologies

- **Node.js:** JavaScript runtime environment for server-side logic.
- **Express.js:** Web framework for building RESTful APIs.
- **Mongoose:** ODM (Object Data Modeling) library for MongoDB schema management.
- **JWT (JSON Web Token):** Authentication and secure session handling.
- **bcrypt:** Secure password hashing.

### 7.4. Database Technology

- **MongoDB:** NoSQL document-oriented database used for storing users, mentorship records, communities, and interaction logs.

### 7.5. Development and Testing Tools

- **Postman:** API testing and validation.
- **Git and GitHub:** Version control and collaborative development.
- **VS Code:** Integrated development environment.

### 7.6. Architecture Pattern

The system follows a three-tier architecture:

- Presentation Layer (React Frontend)
- Application Layer (Node.js + Express Backend)
- Data Layer (MongoDB Database)

This architecture ensures modularity, scalability, and separation of concerns.

## 8. Project Plan and Timeline

### 8.1. Work Completed in Phase I

The following work has been completed in the current semester (Phase I):

- Detailed problem analysis and requirement gathering.

- System architecture design and modular planning.
- MongoDB schema design for:
  - Users
  - Mentorships
  - Communities
  - Posts and Comments
  - Messages
  - Interaction Logs
  - Notifications
- Implementation of institute-based authentication.
- Role-Based Access Control (Junior, Senior, Admin).
- Structured student profile and portfolio module.
- Mentorship lifecycle implementation (Request $\rightarrow$ Accept $\rightarrow$ Active).
- Basic mentor search and filtering based on skills and department.
- Initial version of Mentor Recommendation Engine (rule-based weighted scoring).
- Community module (public/private communities, posts, comments).
- One-to-one messaging structure.
- RESTful API implementation using Express.
- Backend integration with MongoDB including indexing for performance.

### 8.2. Planned Work for Next Semester (Phase II)

The following modules are planned for implementation in the next semester:

- Enhancement of Mentor Recommendation Engine using similarity-based and ML techniques.
- Student clustering using K-Means algorithm for interest-based grouping.
- Association Rule Mining to identify frequently co-occurring domains and skills.
- Engagement trend analysis using timestamped interaction logs.
- Development of Analytics Dashboard with visualizations for:

  - Mentor activity
  - Community engagement
  - Domain popularity
  - Interaction frequency

- Implementation of performance optimization and indexing strategies.
- UI/UX refinement and feature stabilization.
- Deployment preparation and scalability testing.

## 9. References

## References